Vectors

$V = \overrightarrow{AB}$ ⟶ ends at B.

↳ starts at A

Vectors given: Symbols v, w, x

Numbers (Scalars): k, a, b

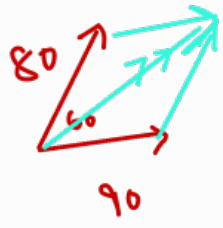
Equal vectors if they have same direc + mag

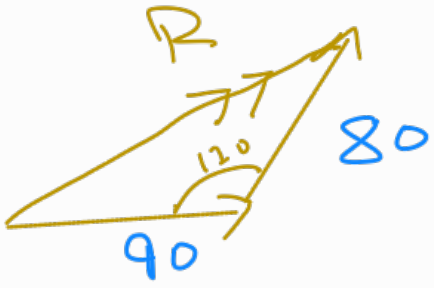
V = w

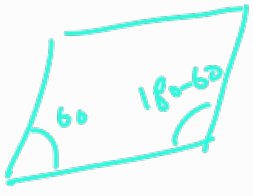
Zero vector

length = 0

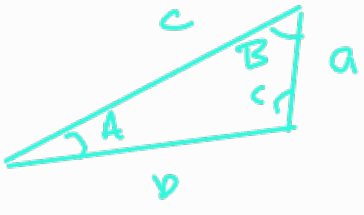
(Vector addition) Parallelogram







adjacent angles are x + y = 180



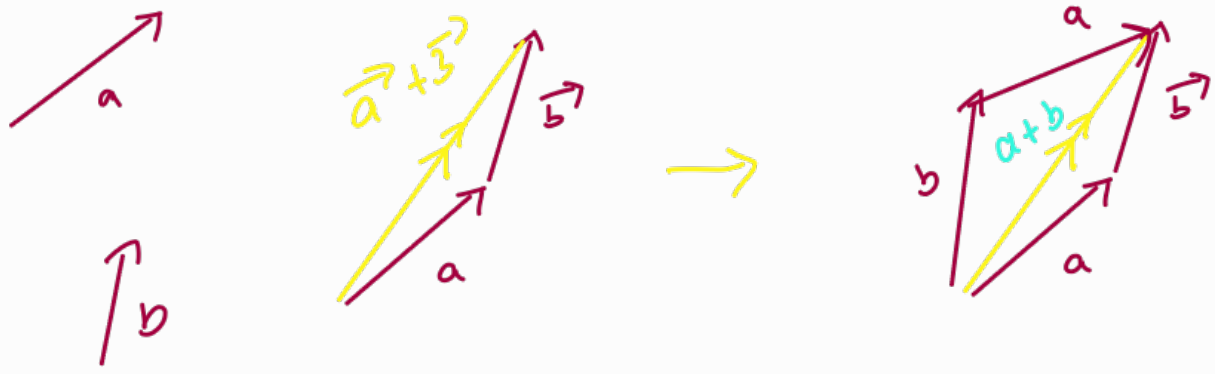
$c^2 = a^2 + b^2 - 2ab\cos C$

↓ angle

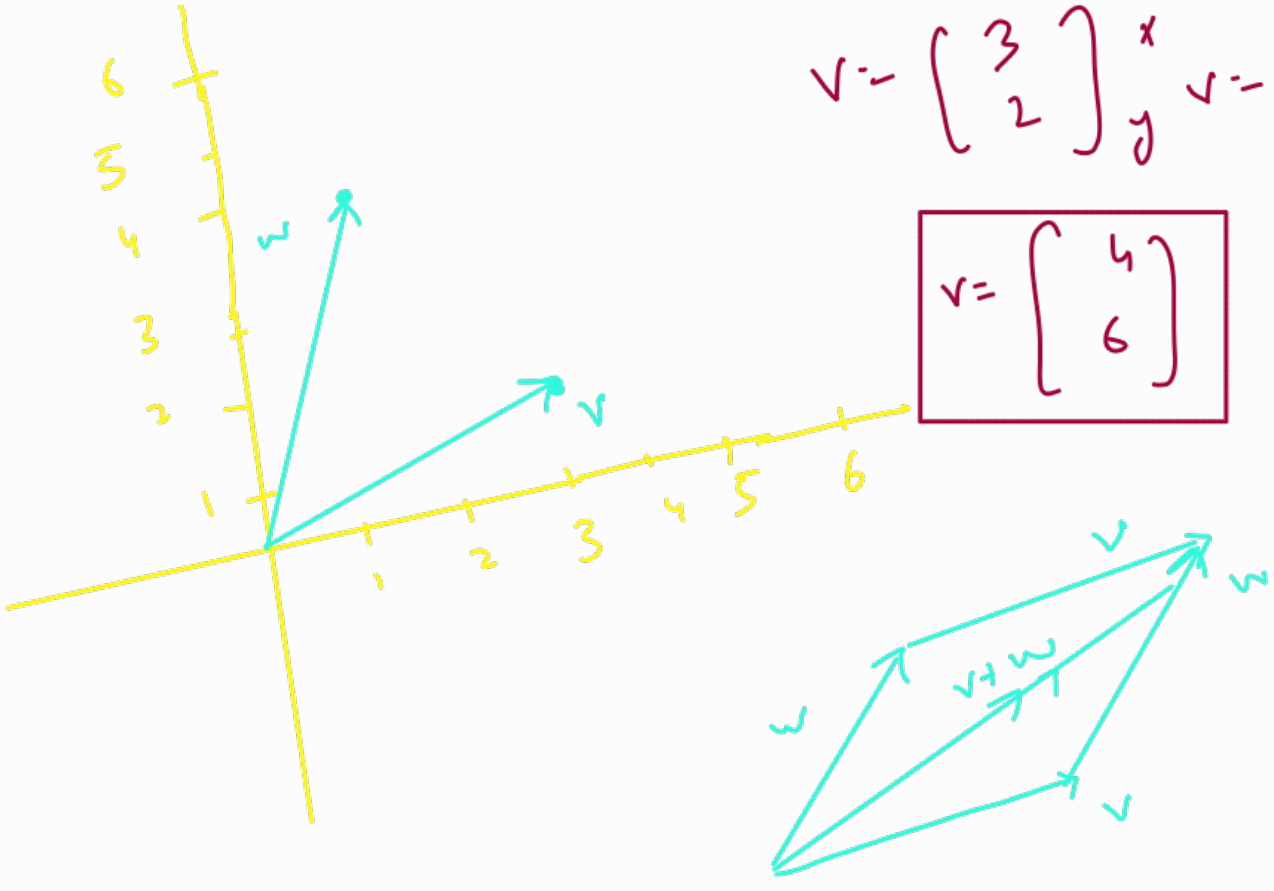
$R^2 = (90)^2 + (80)^2 - 2(90)(80)\cos(120)$

$R = 147.3N$

$\vec{a} + \vec{b}$



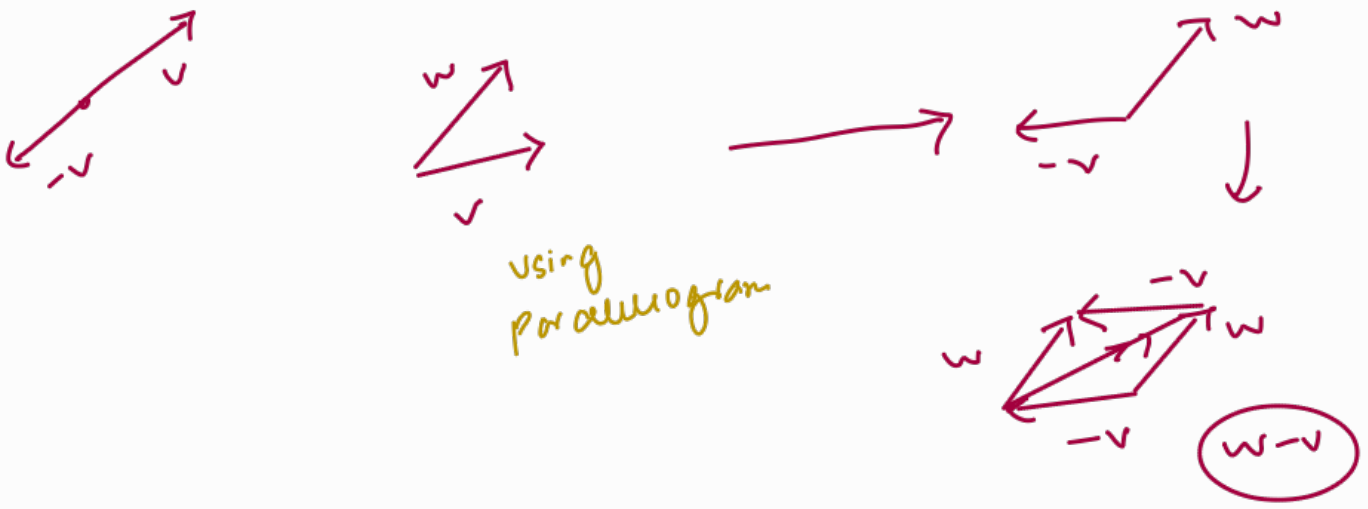
Triangle law → (Just head to tail Rule)



$$v = \begin{bmatrix} 3 \\ 2 \end{bmatrix}_{y}^{x} \quad v = \begin{bmatrix} 1 \\ 4 \end{bmatrix}_{y}$$

$$\boxed{v = \begin{bmatrix} 4 \\ 6 \end{bmatrix}}$$

neg vector → same length, opposite direction.



Scalar multiplication → positive k → stretch/shrink arrow in same direc

↳ negative k → stretch/shrink and flip the arrow

Stretch	$\|k\| > 1$
Shrink	$\|k\| > 1$

Colinear → two vectors lie on the same line. (same s.p)

Parallel → same direc

$u + (v + w) = (u + v) + w$

2D $v = (x_1, y_1)$

3D $v = (x_1, y_1, z_1)$

if $v_1 = (2,3,5)$ $w_2 = (2,3,5)$

$v_1 = w_2$

A vector is just the coordinates of its tip if it starts at the origin.

vector component: final / terminal - initial

ordered n tple → list of ordered numbers

$n=2$ $(3,5)$

$n=3$ $(1,-2,7)$

$n=4$ $(0,4,9,-1)$

$(2,3) \neq (3,2)$

~ Space $\mathbb{R}^n$

$\mathbb{R}^2 \rightarrow (x,y) \rightarrow (2,3), (-1,4)$

$\mathbb{R}^3 \rightarrow (x,y,z) \rightarrow (1,2,3)\ldots$

$v = (1,-3,2)$ $w = (4,2,1)$

i) $v - w = (1-4, -3-2, 2-1) \rightarrow (-3,-5,1)$

---

forms to represent

$v = (2,-1,5)$ $v = [2 \ -1 \ 5]$

$v = \begin{bmatrix} 2 \\ -1 \\ 5 \end{bmatrix}$ → v is in $\mathbb{R}^3$

---

norm of vector / magnitude

$v = (v_1, v_2, \ldots v_n)$

→ Square each component
→ Add them up
→ Take square root

$$\|v\| = \sqrt{v_1^2 + v_2^2 + v_3^2 \cdots v_n^2}$$

↳ dis from the origin to the head of vector

(a) $\|v\| > 0$

(b) $\|v\| = 0$ if and only if $v = 0$

↳ $\|kv\| = |k|\,\|v\|$ multiplying a vector by scalar its length gets multiplied by absolute of that scalar.

Unit vector → vector with length → 1 in same direction.

$$u = \frac{v}{\|v\|}$$

i) Two vectors are in same direc if one is pos multiple of another

$(2, -4, 1/8) \leftrightarrow (1, -2, 1/2, 4)$

ii) Two vectors are in opposite direction if one is neg multiple

$(2, -4, 1/8)$, $(-1, 2, -1/2, -4)$

(Standard unit vectors)

i, J, K

$i \to (1,0)$
$J \to (0,1)$
} $\mathbb{R}^2$

$\mathbb{R}^3$

$i \to (1,0,0)$
$J \to (0,1,0)$
$K \to (0,0,1)$

$i \to$ x-axis $\quad j \to$ y-axis $\quad k \to$ z axis

e.g $v = (2, -1, 5)$ $\quad \boxed{2i - 1j + 5k}$

(For higher dimensions) use $e_1, e_2 \cdots$

$7, 3, -4, 5 \to \boxed{7e_1 + 3e_2 - 4e_3 + 5e_4}$

Distance b/w two points in $\mathbb{R}^n$.

$u = (u_1, u_2, u_3 \cdots) \qquad v = (v_1, v_2, v_3 \cdots)$

norm

$$\|u - v\| = \sqrt{(u_1 - v_1)^2 + (u_2 - v_2)^2 \cdots}$$

(Dot product) (multiplying two vectors)

$$\cos\theta = \frac{u \cdot v}{\|u\| \|v\|}$$

θ: angle b/w two vectors

$A = 2i + 3j + 4k \qquad B = 7i + 5j + k$

$A \cdot B = (2 \cdot 7) + (5 \cdot 3) + (4 \cdot 1) = \boxed{33}$

$\theta = 90^\circ$ Two vectors are orthogonal / perpendicular $\quad u \cdot v = 0$

$\theta = 0^\circ$ parallel $\quad u \cdot v > 0$

$\theta = 180^\circ$ Anti parallel $\quad u \cdot v < 0$

(calculating dot products using components)

($u \cdot v$ for $\mathbb{R}^4$) $u = (-1, 3, 5, 7)$ $v = (-3, -4, 1, 0)$

$u \cdot v = (-1 \times -3,\ 3 \times -4,\ 5 \times 1,\ 7 \times 0)$

$= \boxed{(3 - 12 + 5 + 0) = -4}$

find $(u - 2v) \cdot (3u + 4v)$

$u(3u + 4v) - 2v(3u + 4v)$

$3u^2 + 4u \cdot v - 6u \cdot v + 8v^2$

$\boxed{3u^2 - 2u \cdot v + 8v^2}$

(Cauchy - Schwarz Inequality)

$|u \cdot v| \leq \|u\| \|v\|$

norm of

Dot product of two vectors is never larger than product of there lengths.



Rules

$\|u + v\| \leq \|u\| + \|v\|$ Triangle Rule for vectors

$d(u, v) \leq d(u, w) + d(w, v)$ Triangle Rule for distance

$d(u, v)$ — distance b/w u and v

$d(u, w)$ — distance from u to w

$d(w, v)$ — dis from w to v

going directly from u to v is shorter.

Dot product of column vectors

$u \cdot v = u^T v$ → in matrix notation

$$u = \begin{bmatrix} 1 \\ -3 \\ 5 \end{bmatrix} \quad v = \begin{bmatrix} 5 \\ 4 \\ 0 \end{bmatrix}$$

$u \cdot v = 5 - 12 + 0$

$= \boxed{-7}$

Use $u^T v$ only if (column vector · row vector)

$Au \cdot v = u \cdot A^T v$

$u \cdot Av = A^T u \cdot v$

Note: Matrix multiplication is just dot product

$$A = \begin{bmatrix} x & x \\ y & y \end{bmatrix} \qquad \begin{pmatrix} r_1 \cdot c_1 & r_1 \cdot c_2 \\ r_2 \cdot c_1 & r_2 \cdot c_2 \end{pmatrix}$$

$$B = \begin{bmatrix} + & + \\ - & - \end{bmatrix}$$

$AB =$

Orthogonal vectors are at 90° to each other and their dot product $= 0$

→ Zero vector is orthogonal to every vector.

(Equation of line in 2D)

$$a(x - x_0) + b(y - y_0) = 0$$

(x_0, y_0) known point

a, b → normal vector.

this is a vector $(x - x_0, y - y_0)$

(Equation of plane in 3D)

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$$

x_0, y_0, z_0 → known point

a, b, c → normal vector to plane

general form

→ Line in 2D $ax + by + c = 0$

$n = (a, b)$

$2x + y - 3 = 0$

$(2, 1)$ → normal vector.

→ Plane in 3D $ax + by + cz + d = 0$

$n = (a, b, c)$

$2x - y + 3z = 0$

(a) $(2, -1, 3) \cdot (x, y, z) = 0$

(b) normal vector $(2, -1, 3)$

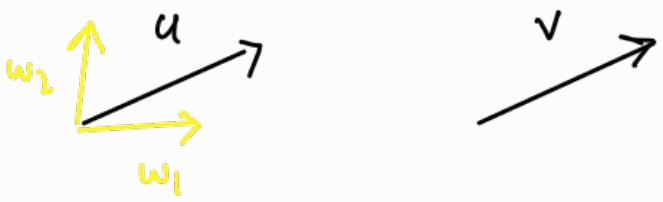
(c) Check if point $(1, -1, 0)$ lies on plane.

$2(1) - (-1) + 3(0) = 0$

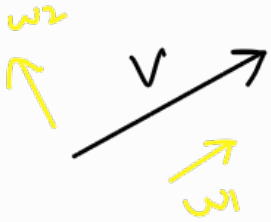
$2 + 1 = 0$

$3 \neq 0$ Doesn't lie on plane

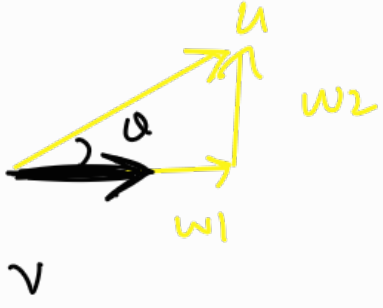
dot product :: of a normal vec and another vector tells you how much that vector goes along the direction of the normal.



Lets say we break u into its components and then compare with vector v.



$w_1 \parallel v$ $\quad$ $w_2 \perp v$



$w_1 = \text{Projection}_v u$

w_1 is projection of vector u on v

w_2 = component of vector u that is orthogonal to vector v.

$$\boxed{w_2 = u - w_1}$$

$$w_1 = \left(\frac{u \cdot v}{\|v\|^2} \right) v$$

$u = (2, -1, 3)$

$a = (4, -1, 2)$

find vector component of u along a and the vector component of u $\perp$ to a.

$w_2 = u - w_1$

$$w_1 = \left(\frac{u \cdot v}{\|v\|^2}\right) v$$

$$w_1 = \left[\frac{(2 \times 4) + (-1 \times -1) + (3 \times 2)}{\sqrt{16 + 1 + 4}}\right] (4, -1, 2)$$

$$w_1 = \left(\frac{8 + 1 + 6}{(\sqrt{21})^2}\right) (4, -1, 2)$$

$$w_1 = \left(\frac{15}{21}\right) (4, -1, 2) \Rightarrow \frac{20}{7}, \frac{-5}{7}, \frac{10}{7}$$

$w_2 = u - w_1$

$w_2 = 2 - \frac{20}{7}, \quad -1 + \frac{5}{7}, \quad 3 - \frac{11}{7}$

norm of projection:

$$= \boxed{\|u\| \cos\theta}$$

$$w_1 = \left(\frac{u \cdot a}{\|a\|^2}\right) a$$

e.g $u = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$ $a = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

$$\|u\| = \sqrt{(3)^2 + (4)^2}$$
$$= \sqrt{9+16}$$
$$= 5$$

$$\cos\theta = \frac{u \cdot v}{\|u\| \|v\|}$$

$$\|v\| = \sqrt{1 + 4}$$
$$= \sqrt{5}$$

$$\cos\theta = \frac{(3 \cdot 1) + (4 \cdot 2)}{(5)\sqrt{5}}$$

$$\cos\theta = \frac{11}{5\sqrt{5}}$$

$$\not{5} \frac{11}{\not{5}\sqrt{5}} = \boxed{\frac{11}{\sqrt{5}}}$$ norm of projection

---

Dis b/w point (x_0, y_0) and line $ax + by + c = 0$

$$D = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$$

Distance b/w point and plane in $\mathbb{R}^3$

$$a(x-x_0) + b(y-y_0) + c(z-z_0) + d = 0$$

$$P_0 (x_0, y_0, z_0)$$

$$D = \frac{|ax_0 + by_0 + cz_0 + d|}{\sqrt{a^2 + b^2 + c^2}}$$

→ normal vectors

TO check if planes ($\mathbb{R}^3$) are parallel

↳ check if normal vectors are parallel

$x + 2y - 2z - 3 = 0$

$2x + 4y - 4z - 7 = 0$

parallel

$(1, 2, -2)$

$(2, 4, -4)$

—

TO check if lines are parallel ($\mathbb{R}^2$)

$n \rightarrow 2n$

Slopes are equal

Line in $\mathbb{R}^4$

$r(t) = P_0 + tv$ → direction vector

↓

Point on a line

(P_1, P_2, P_3, P_4)

Plane in $\mathbb{R}^4$

$r = P_0 + sv_1 + tv_2$

↓

Point on Plane

$\longleftrightarrow$

$n = n_0 + tv$ → direction vector ← by

↑ how far can I move in the direc specified

↓

Starting point

eq of Straight line in $\mathbb{R}^3$

Starts at $(1,2)$

Direc: $(3,1)$

$n = (1,2) + t(3,1)$

$\boxed{n = 1 + 3t}$

$\boxed{y = 2 + t}$

$n = tv$ → line passes through origin

$n = n_0 + t_1v_1 + t_2v_2$ eq of plane in $\mathbb{R}^3$

v_1, v_2 → direction vectors

t_1, t_2 → no to control direc

$n = t_1v_1 + t_2v_2$ → passing through Origin

Vector forms

$x = x_0 + tv$ Line equation in $\mathbb{R}^n$

$x = x_0 + t_1 v_1 + t_2 v_2$ plane equation in $\mathbb{R}^n$

$(x, y, z) = (1, 2, 3) + t(4, -1, 2)$

$$\begin{aligned} x &= 1 + 4t \\ y &= 2 - t \\ z &= 3 + 2t \end{aligned}$$

parametric form of line eq.

$x - y + 2z = 5$ → plane

$y = t \quad z = s$

$x - t + 2s = 5$

$x = 5 + t - 2s$

$(x, y, z) = (5 + t - 2s, t, s)$

finding S.P

$y, z = 0$

$x = 5 + 0 - 2(0)$

$x = 5.$

$(x, y, z) = (5, 0, 0)$

Writing in vector form

$x = x_0 + tv_1 + tv_2$

$(x, y, z) = (5, 0, 0) + (1, 1, 0)t_1 + (-2, 0, 1)t_2$

Suppose $t_1 = 0$ and fix $t_2 = 0$

finding

$(5 + 0 - 2(0), 0, 0)$

$(5, 0, 0)$

now Suppose $t_1 = 1$ fix $t_2 = 0$

$(5 + 1 - 2(0), 1, 0)$ $(6, 1, 0)$

Diff $= (6,1,0) - (5,0,0)$

$\rightarrow (1,1,0) \rightarrow$ value of v_1.

Extra

Use of formula $Au \cdot v = u \cdot A^T v$ (moving A to other side of dot turns it into A^T)

$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ $u = \begin{bmatrix} 5 \\ 6 \end{bmatrix}$ $v = \begin{bmatrix} 7 \\ 8 \end{bmatrix}$

$u \cdot Av = A^T u \cdot v$

Save for this

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}\begin{bmatrix} 5 \\ 6 \end{bmatrix} = \begin{matrix} (1,2) \\ (3,4) \end{matrix} \begin{bmatrix} 15, 6) \\ 17 \\ 39 \end{bmatrix}$$

$$\begin{bmatrix} 7 \\ 8 \end{bmatrix} \cdot \begin{bmatrix} 17 \\ 34 \end{bmatrix} = 7(17) + 8(39) = \boxed{431}$$

$u \cdot A^T v$ $\quad \begin{bmatrix} 5 \\ 6 \end{bmatrix} \cdot \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} 7 \\ 8 \end{bmatrix} = \underline{431}$

Pythagorous theorem in $\mathbb{R}^n$

$$\|u+v\|^2 = \|u\|^2 + \|v\|^2$$

Points from Assign

1) To check if two planes are $\perp$, t/e dot product of their normal vectors.

2) Find distance b/w two parallel planes

use P_2 → arbitary point

Assgn 2 3.4 Q16, 8

equation of line in 3D

$r = r_0 + td$

→ direction vector

Ex 3.4 Line in 2D or 3D

$x = x_0 + tv$

- x_0: point that line goes through
- t: parameter
- v: direc vector

$x = (1,2) + t(3,1)$

$t \to 0 \to (1,2)$

$t \to 1 \to (4,3)$

$t \to -1 \to (-2,1)$

All these points lie on the same line that goes through $(1,2)$ in direction of $(3,1)$

Plane in 3D $\quad x = x_0 + t_1 v_1 + t_2 v_2$

(Yes are the vector forms of line and plane equations)

3.4 geometry of linear Systems: → describing eq in vector form and parametric form.

$$\vec{x} = x_0 + tv$$

(vector form of equation of a line)

↓

Parametric form

$$\vec{x} = (-9, 3, 4) + t(-1, 6, 0)$$

(x, y, z) ← vector x components

$$x = -9 - t$$
$$y = 3 + 6t$$
$$z = 4$$

---

given line $(x, y, z) = (4t, 7, 4 + 3t)$ find a point on the line and a vector parallel to the line

$$x = 4t$$
$$y = 7$$
$$z = 4 + 3t$$

e.g $t = 2$

$x = 8$, $y = 7$, $z = 10$

is a point on the line

Parallel → direction vector same

$$\vec{v} = (4, 0, 3)$$

Vector equation of plane

$$\vec{x} = \vec{x_0} + t_1\vec{v_1} + t_2\vec{v_2}$$

$\vec{x}$ → (x, y, z)

e.g. find the parametric form of the plane in $\mathbb{R}^3$

$$2x + 3y + 5z = 8$$

In parametric form the coefficients are the direction vectors $\vec{v_1}$, $\vec{v_2}$.

let $y = t_1$ $z = t_2$

$$y = t_1 \square + t_2 \square + c$$

$$2x = 8 - 3y - 5z$$

$$2x = -3t_1 - 5t_2 + 8$$

$$x = -\frac{3}{2}t_1 - \frac{5}{2}t_2 + 4$$

$$y = t_1 \quad z = t_2$$

Parametric form.

3.5 cross product:

$u = (u_1, u_2, u_3)$ $v = (v_1, v_2, v_3)$

↓

x, y, z components of u

$u \times v$ → gives another vector that is perpendicular to both u and v, Length depends on angle b/w u and v and direction is found by RHR

$$u \times v = \begin{vmatrix} i & j & k \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix} = i \begin{vmatrix} u_2 & u_3 \\ v_2 & v_3 \end{vmatrix} - j \begin{vmatrix} u_1 & u_3 \\ v_1 & v_3 \end{vmatrix} + k \begin{vmatrix} u_1 & u_2 \\ v_1 & v_2 \end{vmatrix}$$

$(u_2v_3 - u_3v_2)$

e.g $u \times v$ $u = \langle 1, 2, -2 \rangle$

$v = \langle 3, 0, 1 \rangle$

$$u \times v = \begin{vmatrix} i & j & k \\ 1 & 2 & -2 \\ 3 & 0 & 1 \end{vmatrix} = i \begin{vmatrix} 2 & -2 \\ 0 & 1 \end{vmatrix} - j \begin{vmatrix} 1 & -2 \\ 3 & 1 \end{vmatrix} + k \begin{vmatrix} 1 & 2 \\ 3 & 0 \end{vmatrix}$$

$= \boxed{2i - 7j - 6k}$

Relationship b/w cross product and dot product

$u \cdot (u \times v) = 0$

↓

gives another vector

when two vectors are perpendicular then their dot product is zero

$v \cdot (u \times v) = 0$

$$\|u \times v\|^2 = \|u\|^2 \|v\|^2 - (u \cdot v)^2$$

$$u \times (v \times w) = (u \cdot w)v - (u \cdot v)w$$

$$(u \times v) \times w = (u \cdot w)v - (v \cdot w)u$$

} Properties

$$\boxed{i \times j = k}$$

Properties of cross product

$$u \times v = -(v \times u)$$

$$u \times (v + w) = u \times v + u \times w$$

$$k(u \times v) = ku \times v = u \times kv$$

$$u \times 0 = 0$$

$$u \times u = 0$$

(geometric interpretation) Area of parallelogram $= |u \times v|$

c.j find the area of triangle determined by the points

$P_1\ (3,2,0)$, $P_2\ (-1,0,2)$ $P_3\ (0,4,3)$

Area of triangle: $\frac{1}{2} \times b \times h$ → Area of parallelogram

$$\overrightarrow{P_1P_2} = (-1,0,2) - (3,2,0) = (-3,-2,2)$$

$$\overrightarrow{P_1P_3} = (0,4,3) - (2,2,0) = (-2,2,3)$$

$$(-3,-2,2) \times (-2,2,3)$$

$u_1\ u_2\ u_3 \quad v_1\ v_2\ v_3$

$$\begin{vmatrix} i & J & k \\ -3 & -2 & 2 \\ -2 & 2 & 3 \end{vmatrix}$$

$$i\begin{vmatrix} -2 & 2 \\ 2 & 3 \end{vmatrix} - J\begin{vmatrix} -3 & 2 \\ -2 & 3 \end{vmatrix} - k\begin{vmatrix} -3 & -2 \\ -2 & 2 \end{vmatrix}$$

$$(-6-4)i - J(-9+4) + k(-6-4)$$

$$(-10i + 5J - 10k)$$

$$\sqrt{(10)^2 + (5)^2 + (-10)^2} = \left(\frac{15}{2}\right) -$$

Scalar triple product:

$u \cdot (v \times w)$ → gives vol of parallelepiped formed by 3 vectors (u, v, w)

$u = (u_1, u_2, u_3)$

$v = (v_1, v_2, v_3)$

$w = (w_1, w_2, w_3)$

$$u \cdot (v \times w) = \begin{vmatrix} u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{vmatrix} = u_1 \begin{vmatrix} v_2 & v_3 \\ w_2 & w_3 \end{vmatrix}$$

$$- u_2 \begin{vmatrix} v_1 & v_3 \\ w_1 & w_3 \end{vmatrix} + u_3 \begin{vmatrix} v_1 & v_2 \\ w_1 & w_2 \end{vmatrix}$$

$u(v \times w) = 0$ ∴ (vectors lies on the same plane no thickness. hence no vol)

$u(v \times w) \neq 0$ 3D